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Inventor(s)

Jacob; Dylan M. et al.

VERTICAL-FEED PERSONAL HYDRATION SYSTEM

Abstract

Systems and methods for a vertical-feed personal hydration system are provided. In some embodiments, a personal hydration system is provided that facilitates drainage of fluid from a fluid delivery tube back into the fluid reservoir when a user has completed taking a drink from a mouthpiece. The personal hydration system may comprise a venting cap assembly comprising a straw that extends down into a reservoir of a hydration vessel, and a fluid delivery tube that extends between the venting cap assembly and a valve operated mouthpiece. The venting cap assembly may include a quick release hose connector used to quickly separate and re-attach the fluid delivery tube from the hydration vessel. The hydration vessel may comprise a stainless steel alloy material, and/or fabricated using other alloys or materials such as a titanium alloy, a ceramic material, or other composite materials. The hydration vessel may comprise an insulated vessel.

Inventors: Jacob; Dylan M. (Denver, CO), Westerberg; Jordan (Denver, CO)

Applicant: Brumate, Inc. (Denver, CO)

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Background/Summary

BACKGROUND

[0001] A personal hydration system, often referred to as a hydration pack, is a portable system designed to provide a person with a convenient way for a user to remain hydrated during outdoor activities such as hiking, biking, running, or camping, among other activities. A hydration system may include a bladder, or other form of reservoir, for holding a hydrating fluid (e.g., water), and a hose or tube connecting the reservoir to a mouthpiece used for drinking. The hydration system may be worn by the user, such as in a pouch or a backpack, with the mouthpiece routed over their shoulder, providing the user with a hands-free way to drink from the reservoir while on the move.

SUMMARY

[0002] This summary is provided to introduce a selection of concepts in a simplified form that are further described below in the detailed description. This summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used in isolation as an aid in determining the scope of the claimed subject matter.

[0003] One or more of the embodiments presented in this disclosure, among other thing, provide for a vertical-feed based personal hydration system that facilitates the drainage of fluid from a fluid delivery tube and back into the fluid reservoir when the user has completed taking a drink from a mouthpiece. The personal hydration system may comprise a venting cap assembly comprising a straw that that extends down into a reservoir of a hydration vessel, and a fluid delivery tube that extends between the venting cap assembly and a valve operated mouthpiece. The venting cap assembly may include a quick release hose connector used to quickly separate and re-attach the fluid delivery tube from the hydration vessel. The hydration vessel may comprise a stainless steel alloy material, and/or fabricated using other alloys or materials such as a titanium alloy, a ceramic material, or other composite materials. The hydration vessel may comprise an insulated vessel. The personal hydration system may function as a vertical-feed system in that hydrating fluids are drawn by the user from the reservoir by opening the valve operated mouthpiece, and applying a suction to a drinking port of the mouthpiece. When the user has completed taking a drink from the hydration system, they can cease applying suction to the mouthpiece and leave the mouthpiece open until fluid remaining in the fluid delivery tube drains back into the reservoir. As such, the non-dispensed fluid does not remain in the fluid delivery tube, where it may be subject to undesired heating and/or freezing due to environmental conditions, and/or absorbing undesired tastes from remaining in the fluid delivery tube.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] Aspects of the present disclosure are described in detail herein with reference to the attached Figures, which are intended to be exemplary and non-limiting, wherein:

[0005] FIG. 1 is a diagram illustrating an example vertical-feed based personal hydration system, in accordance with an embodiment of the present disclosure;

[0006] FIG. 2 is a cross sectional illustration an example of an hydration vessel and venting cap assembly, in accordance with an embodiment of the present disclosure;

[0007] FIG. 3 is a cross sectional illustration further detailing features of an example venting cap assembly, in accordance with an embodiment of the present disclosure;

[0008] FIG. 4 is a drawing illustrating an alternate view of an example venting cap assembly, in accordance with an embodiment of the present disclosure;

[0009] FIG. 5 is a drawing illustrating a partially exploded view of an example venting cap

assembly, in accordance with an embodiment of the present disclosure;

[0010] FIG. 6 is a diagram illustrating an example of a venting cap assembly in combination with a hydration vessel, in accordance with an embodiment of the present disclosure;

[0011] FIGS. 7A and 7B are diagrams illustrating an example of a mouthpiece comprising a flow control valve, in accordance with an embodiment of the present disclosure; and

[0012] FIG. 8 is a cross-section drawing of the hydration vessel with a vertical-feed based venting cap assembly operating as a vertical-feed system, in accordance with an embodiment of the present disclosure.

DETAILED DESCRIPTION

[0013] In the following detailed description, reference is made to the accompanying drawings that form a part hereof, and in which is shown specific illustrative example embodiments in which the embodiments may be practiced. These embodiments are described in sufficient detail to enable those skilled in the art to practice the embodiments, and it is to be understood that other embodiments may be utilized, and that logical, mechanical, and electrical changes may be made without departing from the scope of the present disclosure. The following detailed description is, therefore, not to be taken in a limiting sense.

[0014] Personal hydration systems, particularly those worn in conjunction with backpacks and/or slings, are primarily gravity fed systems that rely on the force of gravity to deliver water from a reservoir (e.g., a bladder) to a drinking nozzle. For example, a hose may be attached to a bottom the reservoir, and water (or other fluid) flows down through the hose to the drinking nozzle for consumption by the user. The drinking nozzle may comprise a flow control valve in the form of a bite valve. The user may open the drinking nozzle to initiate flow through the hose and dispense water into the mouth of the user by biting down on the bite valve. A difference in elevation between the reservoir and the drinking nozzle (with the reservoir at a higher elevation relative to the drinking nozzle) permits gravity to provide the force for transporting the water through the hose.

[0015] However, several problems may occur when using gravity fed personal hydration systems with respect to stagnate fluids remaining in the hose between drinking sessions. For example, when the user finishes taking a drink from the drinking nozzle, they may release their bite on the bite valve to close the drinking nozzle and stop the flow of water. Closing of the drinking nozzle seals that end of the hose, essentially forming a backpressure in the hose that prevents the water from draining. Further, with a gravity fed system, while gravity provides the force that motivates the flow of water from the bottom of the reservoir, through the hose, and out from the drinking nozzle, there is no counter-force to motivate water to drain from the hose back into the reservoir when drinking is completed. As such, during warm environmental conditions (such as where the hydration system is being used in conjunction with activities such as hiking, biking, and/or warm weather camping) the water in the hose may become very warm such that when the user operates the drinking nozzle to quench their thirst, the drinking nozzle undesirably dispenses warm water directly into the user's mouth. Moreover, in such conditions where the water in the hose may be warmed due to warm ambient conditions, undesirable flavor characteristics may leach from the hose into the water-substantially degrading the experience of drinking from the hydration system.

Conversely, in cold ambient conditions (such as where the hydration system is being used in conjunction with skiing, snowboarding, snowshoeing and/or cold weather camping) the water in the hose may freeze, limiting and/or blocking the flow of water to the drinking nozzle, and/or damaging components of the hydration system due to ice expanding within the system.

[0016] In contrast to such prior hydration systems, embodiments of the present disclosure, among other things, provide for a vertical-feed based personal hydration system that facilitates the drainage of fluid from a fluid delivery tube and back into the fluid reservoir when the user has completed taking a drink from a mouthpiece. More specifically, in some embodiments, a personal hydration system may comprise a venting cap assembly comprising a straw that extends down into a reservoir of a hydration vessel, and a fluid delivery tube that extends between the venting cap

assembly and a valve operated mouthpiece. The straw that extends into the hydration vessel may be detachable from the venting cap assembly. The hydration vessel may comprise a stainless steel alloy material, and/or fabricated using other alloys or materials such as a titanium alloy, a ceramic material, or other composite materials. The hydration vessel may comprise an insulated vessel. For example, the body of the hydration vessel may comprise a double-walled housing that includes an outer wall and an inner wall separated by an insulating medium. In some embodiments, the insulating medium comprises a vacuum formed between the outer wall and the inner wall. In some embodiments, the insulating medium **213** may comprise an insulating foam or other material having a low thermal conductivity (e.g., Styrofoam).

[0017] The personal hydration system may function as a vertical-feed system in that hydrating fluids (e.g., water) are drawn by the user from the reservoir by opening the valve operated mouthpiece, and applying a suction to a drinking port of the mouthpiece. With the suction applied, the fluid is drawn vertically up the straw of the venting cap assembly, through the venting cap assembly and fluid delivery tube, and dispensed from the drinking port. The venting cap assembly comprises an equalization valve that opens to permits outside air to enter the hydration vessel in response to the user drawing fluid from the hydration vessel—thereby avoiding a buildup of negative pressure inside the hydration vessel. Moreover, the venting cap assembly may include a quick release hose connector used to quickly separate and re-attach the fluid delivery tube from the hydration vessel.

[0018] The mouthpiece may comprise a position-latching flow control valve. That is, the user applies an actuation force to open the mouthpiece, and an actuation force to close the mouthpiece. The state of the mouthpiece flow control valve (e.g., an open state or a closed state) remains latched until a force is applied by the user to switch the mouthpiece to the other state. When the user has completed taking a drink from the hydration system, they can cease applying suction to the mouthpiece and leave the mouthpiece open until fluid remaining in the fluid delivery tube drains back into the reservoir. As such, the non-dispensed fluid does not remain in the fluid delivery tube, where it may be subject to undesired heating and/or freezing due to environmental conditions, and/or absorbing undesired tastes from remaining in the fluid delivery tube. In some embodiments, the fluid may be drawn back into the reservoir based on the force of gravity and/or a differential in pressure between atmospheric pressure at the open mouthpiece and the air within the reservoir. Once the fluid has drained from the fluid delivery tube, the user may close the valve of the mouthpiece.

[0019] Referring now to FIG. 1, FIG. 1 is a diagram illustrating a hydration system **100** according to embodiments of the present disclosure. As shown in FIG. 1, the hydration system **100** may include a valve operated mouthpiece **110** coupled to one end of a fluid delivery tube **112**. The opposing end of the fluid delivery tube **112** may, in turn, be coupled to a hydration vessel **130** via a venting cap assembly **120**. In some embodiments, the mouthpiece **110** may comprise a position-latching flow control valve, meaning that the flow control valve may switch between opened and closed states in response to the application of an actuation force. For example, when the user applies an actuation force to a closed mouthpiece **110**, the flow control valve of the mouthpiece **110** opens and remains open until a subsequent force is applied to close the mouthpiece **110**. When the user applies an actuation force to an open mouthpiece **110**, the valve of the mouthpiece **110** closes and remains closed until a subsequent force is applied to open the mouthpiece **110**. Examples of a position-latching valves for mouthpiece **110** may include, but are not limited to, a push-pull valve, a dial-controlled valve, a lever-controlled valve, and/or a button-activated valve. In contrast, a bite valve is an example of a non-position-latching valve because a closed bite valve opens in response to the pressure applied by the force of a user biting on the valve, and automatically re-closes in response to the removal of that pressure. The fluid delivery tube **112** may comprise any type of flexible and/or semi-flexible tubing material that can transport fluid from the hydration vessel **130** in response to a vacuum being drawn from the mouthpiece **110** (e.g., by a user drawing a vacuum

using their mouth to drink from the hydration vessel **130**).

[0020] The hydration vessel **130** may comprise a stainless steel alloy material (e.g., an 18/8, 18/10, 304, or 316, grade stainless steel), and/or fabricated using other alloys or materials such as a titanium alloy, a ceramic material, or other composite materials. Moreover, in some embodiments, hydration vessel **130** may comprise an insulated vessel. That is, the body of the hydration vessel **130** may comprise a first outer wall and a second inner wall, that are separated by a thermal insulator such as a vacuum and/or thermal insulating materials. The hydration vessel **130** may be coupled to the fluid delivery tube **112** by the venting cap assembly **120**. As described greater with respect to FIG. 2, the venting cap assembly **120** may include a quick release hose connector that may be used to quickly detach and re-attach the fluid delivery tube **112** from the hydration vessel **130**. Moreover, the venting cap assembly **120** comprises a cap assembly that includes an equalization valve (such as equalization valve **400** discussed below) that permits outside air to enter the hydration vessel **130** as the user draws fluid from the hydration vessel **130**—thereby avoiding a buildup of negative pressure inside the hydration vessel **130** as the user drinks from the hydration vessel **130**. That is, the venting cap assembly **120** may comprises an equalization valve configured to open in response to a suction applied to the mouthpiece **110**, to pass an equalizing air into the reservoir **218** from an environment external to the hydration vessel **130**. As a result, the amount of effort for the user to draw fluid from the hydration vessel **130** to the mouthpiece **110** will remain constant or nearly constant and not increase due to increasing back-pressure developing in the hydration vessel **130**. After the user has finished drawing from the hydration vessel **130**, they may release the mouthpiece **110** from their mouth. Because the mouthpiece **110** comprises a state-sustaining valve, it may remain in an open state such that the external air-pressure and/or gravity causes the fluid remaining in the fluid delivery tube **112** to drain back into the hydration vessel **130**—thus avoiding a configuration where fluid within the fluid delivery tube **112** freezes or heats-up do to environmental conditions, or otherwise remains stagnant within the fluid delivery tube **112**.

[0021] Referring now to FIG. 2, FIG. 2 is a cross sectional illustration of the hydration vessel **130** and venting cap assembly **120**. Starting with the hydration vessel **130**, as shown in FIG. 2, the hydration vessel **130** comprises a housing **210** that includes an interior volume that defines a reservoir **218** for holding a hydrating fluid. The housing **210** may comprise an insulating housing. For example, the housing **210** may comprise a double-walled housing that includes an outer wall **211** and an inner wall **212** separated by an insulating medium **213**. In some embodiments, the insulating medium **213** comprises a vacuum formed between the outer wall **211** and the inner wall **212**. In some embodiments, the insulating medium **213** may comprise an insulating foam or other material having a low thermal conductivity (e.g., Styrofoam).

[0022] As shown in FIG. 2, in some embodiments the venting cap assembly **120** may comprise a quick release hose connector **230**, a vessel connector **220**, a straw coupler **216**, and/or a straw **215**. As discussed herein, the venting cap assembly **120**, in combination with the fluid delivery tube **112** and mouthpiece **110**, define a vertical-feed hydration system that may be used with a hydration vessel (e.g., such as hydration vessel **130**) that delivers fluid from a reservoir **218** to the mouthpiece **110** for consumption by the user, and that also permits the fluid remaining in the fluid delivery tube **112** to drain back into the hydration vessel **130** between uses, instead of remaining stagnant in the fluid delivery tube **112**. The vessel connector **220** may, among other things, provide an interface for securely fastening the venting cap assembly **120** to the hydration vessel **130**, and supporting the straw **215** within the reservoir **218**. That is, the vessel connector may at least in part function to fasten the venting cap assembly **120** to the hydration vessel **130**.

[0023] As shown in FIG. 2, a proximal end of the straw **215** may connect to the vessel connector **220** via the straw coupler **216** at a position near the top **201** of the reservoir **218** while the distal end of the straw **215** is supported in a position suspended above the bottom **202** of the reservoir **218**. As such, when fluid is drawn from the reservoir **218**, the fluid is vertically fed (e.g., pulled or lifted) from the bottom **202** of the reservoir. As discussed in greater detail below, the vessel connector **220**

further includes an equalization valve that permits air from the surrounding environment to flow into the reservoir **218** while fluid is drawn out from the reservoir **218** through the straw **215**, thus avoiding a buildup of backpressure within the reservoir **218** while fluid is being drawn out. In various embodiments, the straw **215** may be fabricated from materials such as a stainless steel alloy material (e.g., an 18/8, 18/10, 304, or 316, grade stainless steel), and/or fabricated using other alloys or materials such as a titanium alloy, ceramic materials, or other composite materials.

[0024] The venting cap assembly **120** may further comprise a hose connector **230** (e.g., a quick release hose connector or other removable hose connector) that is coupled to an opposing end of the fluid delivery tube **112** from the mouthpiece **110**. The quick release hose connector **230** may couple to the vessel connector **220** via a quick release fastening mechanism (or other fastening mechanism) to facilitate easy disconnection and/or reconnection of the fluid delivery tube **112** and mouthpiece **110** from the hydration vessel **130**. In some embodiments, as discussed with respect to FIG. **3**, the vessel connector **220** comprises a spring activated valve that seals the reservoir **218** when the quick release hose connector **230** is decoupled from the vessel connector **220** to prevent spillage and/or contamination of fluid within the reservoir **218**. Although the venting cap assembly **120** is primarily described herein as used in conjunction with the hydration vessel **130**, such examples are for illustration purposes only and not intended as limiting. Other embodiments may include a venting cap assembly **120** that may be used in combination with other vessels to produce a vertical-feed based hydration system.

[0025] Referring now to FIG. **3**, FIG. **3** is a cross sectional drawing further illustrating features of the venting cap assembly **120**. As shown in FIG. **3**, in some embodiments the vessel connector **220** may comprise a fastening mechanism, such as threads **314** that engage with corresponding threads **312** at the opening of the hydration vessel **130**, in order to secure the venting cap assembly **120** to the hydration vessel **130**. In other embodiments, other types of fasteners may be used to connect the vessel connector **220** to the hydration vessel **130**. As described herein, the venting cap assembly **120** comprises an interior pathway defining an internal flow path between the straw **215** and the fluid delivery tube **112**.

[0026] Within the reservoir **218**, the vessel connector **220** may comprise a port **320** configured to interface with the straw coupler **216** to couple the straw **215** to the venting cap assembly **120**. That is, the straw **215** may comprise a removable straw where the straw **215** may be attached and/or detached from the port **320** using the straw coupler **216**. For example, the port **320** may comprise at least part of a keyed fastening mechanism, while the straw coupler **216** may comprise a corresponding part of the keyed fastening mechanism where one or more pins or other protrusions that engage with the keyed fastener (such as a keyed twist-to-lock fastening mechanism) to fasten the straw coupler **216** and the straw **215** to the vessel connector **220**. Alternatively, in some embodiments, the port **320** may comprise threads that engage with threads of the straw coupler **216** to form a threaded connection. In other embodiments, other types of fasteners may be used to connect the straw coupler **216** and/or straw **215** to the venting cap assembly **120**.

[0027] The quick release hose connector **230** may comprise a hose connector port **344** (e.g., a barbed connector port) to couple to the fluid delivery tube **112**. In some embodiments, the quick release hose connector **230** and vessel connector **220** may comprise corresponding member of a fastener **322** that facilitates coupling and securing the quick release hose connector **230** to the vessel connector **220**. The fastener **322** may include a quick release fastening mechanism, such as a twist-to-lock fastening mechanism **512** where the quick release hose connector **230** comprise one or more pins that engage with the vessel connector **220** in a twist-to-lock fashion to fasten the quick release hose connector **230** to the vessel connector **220**. Alternatively, in some embodiments, the quick release hose connector **230** may comprise threads that engage with threads of the vessel connector **220** to form a threaded fastening mechanism. In other embodiments, other types of fasteners may be used to connect the quick release hose connector **230** to the vessel connector **220**.

[0028] As further illustrated in FIG. **3**, the vessel connector **220** may comprise a reservoir sealing

valve **332** that operates to open and close in response to connecting and disconnecting the quick release hose connector **230**. In some embodiments, the reservoir sealing valve **332** may comprise a spring-loaded poppet valve. The vessel connector **220** may comprise a coil spring **336** positioned around a stem **335** of the reservoir sealing valve **332** that is pre-loaded to apply a force against a stop member **338** coupled to the stem **335**. With the quick release hose connector **230** disconnected from the vessel connector **220**, the force of the coil spring **336** applied to the stop member **338** pulls the reservoir sealing valve **332** into a closed position within a first fluid pathway chamber **330** of the vessel connector **220**, where the fluid pathway chamber **330** has an open flow path to the reservoir **218** via the straw **215**. With the reservoir sealing valve **332** in this closed position, the reservoir **218** is effectively sealed to prevent spillage and/or contamination of fluid within the reservoir **218**. In order to open the reservoir sealing valve **332** in response to an attachment of the quick release hose connector **230**, the quick release hose connector **230** may comprise a valve actuator **340** that functions to apply a force to depress the stem **335** of the reservoir sealing valve **332** and compress the coil spring **336** as the quick release hose connector **230** is attached to the vessel connector **220**. As the stem **335** of the reservoir sealing valve **332** is depressed by the valve actuator **340**, the reservoir sealing valve **332** is pushed into an open position within the first fluid pathway chamber **330**—thus opening a flow path between the first fluid pathway chamber **330** and a second fluid pathway chamber **342** defined within the interior of the quick release hose connector **230**. The second fluid pathway chamber **342**, in turn, is coupled to the hose connector port **344**. As such, in some embodiments, the attachment of the quick release hose connector **230** to the vessel connector **220** opens the reservoir sealing valve **332** creating a continuous flow path for fluid flow between the straw **215** (and/or the reservoir **218**) and the mouthpiece **110**.

[0029] Referring now to FIG. 4, FIG. 4 illustrates an alternate view of the venting cap assembly **120** discussed with respect to FIG. 3, with the quick release hose connector **230** fastened to the vessel connector **220**, and the port **320** of vessel connector **220** coupled to a straw coupler **216** (where straw **215** is not illustrated in order to simplify the drawing). In particular, FIG. 4 depicts the equalization valve **400** of the venting cap assembly **120** discussed herein. In some embodiments, the equalization valve **400** may operate as a one-way valve (or check-valve) that permits air to flow into the reservoir **218** in response to fluid being drawn out from reservoir **218** through the straw **215**, but does not permit fluid from the reservoir **218** to leak out of the reservoir **218** from the equalization valve **400**. In some embodiments, the equalization valve **400** is at least in part located on an exterior wall of the vessel connector **220** that is exposed to the reservoir **218** when the vessel connector **220** is installed into hydration vessel **130**. In some embodiments, the equalization valve **400** may draw in air for equalizing the reservoir **218** from a cavity **410** of the vessel connector **220** that is exposed to the external environment.

[0030] Referring now to FIG. 5, FIG. 5 illustrates an alternate partially exploded view of the venting cap assembly **120** discussed with respect to FIG. 2, with the quick release hose connector **230** and fastening mechanism **322** detached from the vessel connector **220**. The vessel connector **220** may comprise a cavity **410** for receiving the fastening mechanism **322** and quick release hose connector **230**, and insertion of the quick release hose connector **230** depresses the stem **335** of reservoir sealing valve **332** downward to open the reservoir sealing valve **332**. In some embodiments, the quick release hose connector **230** may comprises one or more sealing gaskets **520** that seal against an interior wall **510** of the vessel connector **220** to create a water-tight seal when the quick release hose connector **230** is connected to the vessel connector **220**.

[0031] FIG. 6 illustrates the insertion of the venting cap assembly **120** to the hydration vessel **130** in order to fasten those elements together. The straw **215** may be inserted into the opening of the hydration vessel **130** (e.g., into reservoir **218**) and the venting cap assembly **120** secured by engaging the threads **314** of the venting cap assembly **120** with the threads **312** about the opening of the hydration vessel **130**. In some embodiments, the venting cap assembly **120** may comprise one or more gaskets **610** that seal against a flange **612** of the hydration vessel **130** to form a water-

tight seal between the venting cap assembly **120** and the hydration vessel **130**. As illustrated by FIG. **6**, when the venting cap assembly **120** is installed, the equalization valve **400** may be positioned in alignment with the threads **312** and **314** such that the threads provide an air passage for equalizing air to flow from the equalization valve **400** into the reservoir **218**.

[0032] FIGS. **7A** and **7B** are diagrams illustrating an example mouthpiece **110** in accordance with embodiments of the present disclosure. In this example embodiment, the mouthpiece **110** comprises a push-pull type of position-latching valve. However, embodiments may include other types of mouthpieces other than shown in FIGS. **7A** and **7B**. Other flow control valves type that may be used for mouthpiece **110** may include, but are not limited to, a dial-controlled valve, a lever-controlled valve, a button-activated valve, other position-latching valve types, other valves that facilitate clearing of fluid from the fluid delivery tube **112** back into the hydration vessel **130**. In some embodiments, mouthpiece **110** may at least in part comprise a bite valve.

[0033] As shown in FIG. **7A**, the mouthpiece **110** may include a body **714** that comprises a hose connector **716** (e.g., a barbed connector port) that may be used to couple the mouthpiece **110** to the fluid delivery tube **112**. In some embodiments, the body **714** may include a bend or angle (as shown at **718**) to help orient a drinking port **734** of the mouthpiece **110** towards the mouth of the user. The drinking port **734** may include a port cover **710** that comprises, for example, a plastic, soft silicone, or similar material that is coupled to a valve actuator **712** of the mouthpiece **110**. As described herein, the mouthpiece **110** comprises an interior pathway defining a flow path between the fluid delivery tube **112** and a drinking port **734** of the mouthpiece **110**.

[0034] Referring to FIG. **7B**, the valve actuator **712** may extend into the body **714** of the mouthpiece **110**, and comprise a stopper member **720** that seals against an internal wall **726** of the mouthpiece **110** to define a push-pull valve. In some embodiments the stopper member **720** may include a gasket **724** (e.g., an o-ring) to create a water-tight seal between the stopper member **720** and the internal wall **726** when the push-pull valve is in the closed position. When the stopper member **720** is disengaged from the internal wall **726** (e.g., the push-pull valve is open), an internal channel **730** within the body **714** may freely communicate with an internal channel **732** within valve actuator **712** to complete a flow path (e.g., for air and/or fluids) between the drinking port **734** and the hose connector **716**. When the stopper member **720** is engaged with the internal wall **726** (e.g., the push-pull valve is open), the internal channel **730** within the body is isolated from the internal channel **732** within valve actuator **712** to close the flow path between the drinking port **734** and the hose connector **716**. Operating as a position-latching valve, a closed flow path within the mouthpiece **110** remains closed until the valve actuator **712** is pulled out from the body **714** to open it, and an open flow path within the mouthpiece **110** remains open until the valve actuator **712** is pushed into the body **714** to close it.

[0035] Referring now to FIG. **8**, FIG. **8** is a cross-section drawing of the hydration vessel **130** with a vertical-feed based venting cap assembly **120** operating as a vertical-feed system that may provide hydrating fluid to a user of a personal hydration system **100**. As shown in FIG. **8**, the quick release hose connector **230** is connected to the vessel connector **220**, thus opening reservoir sealing valve **332** to permit fluid flow out from the reservoir **218** and into fluid delivery tube **112** in response to a vacuum drawn from the mouthpiece **110** (e.g., from a user using their mouth to draw the fluid through an open mouthpiece **110**). An application of the negative pressure of a vacuum applied to the mouthpiece **110** overcomes the force of gravity on fluid **810** in the reservoir **218**, and feeds that fluid **810** from the reservoir **218** vertically up through the straw **215**, through the internal pathways within the venting cap assembly **120**, until the fluid reaches the mouthpiece **110**, from which the fluid **810** is dispensed to the user. The pull of fluid **810** into straw **215** causes a negative pressure within reservoir **218**, and a corresponding pressure differential across equalization valve **400** that opens equalization valve **400** to permit and equalizing airflow **820** to flow into reservoir **218** which limits the buildup of negative pressure within reservoir **218**. Limiting this backpressure increases the effectiveness of personal hydration system **100**, and improves user experience. That

is, the user does not need to exert increasingly greater levels of effort to apply suction to extract fluid from the mouthpiece **110** as they are drinking from the reservoir **218** (as would be the case if without equalizing airflow **820**) increasing their ability to consistently judge the suction they need to apply to extract a given unit of fluid from the reservoir **218**. When the user has completed taking a drink from the hydration system **100**, they can cease applying suction to the mouthpiece **110** and leave the mouthpiece **110** open until fluid remaining in the fluid delivery tube **112** drains back into the reservoir **218** of the insulated hydration vessel **130**. As such, the non-dispensed fluid does not remain in the fluid delivery tube **112**, where it may be subject to undesired heating and/or freezing due to environmental conditions, and/or absorbing undesired tastes from remaining in the fluid delivery tube **112**. In some embodiments, the fluid may be drawn back into the reservoir **218** based on the force of gravity and/or a differential in pressure between atmospheric pressure at the open mouthpiece **110** and the air within the reservoir **218**. Once the fluid has drained from the fluid delivery tube **112**, the user may close the valve of the mouthpiece **110**.

[0036] Various modifications and different embodiments are described herein in detail with reference to the accompanying drawings so that those skilled in the art can carry out the disclosure. It should be understood, however, that the present disclosure is not intended to be limited to the specific embodiments, but the present disclosure includes modifications, equivalents or replacements that fall within the spirit and scope of the disclosure as defined in the following claims.

[0037] The terminology used herein is for the purpose of describing specific embodiments only and is not intended to limit the scope of the disclosure. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. In the disclosure, terms such as “comprises”, “includes”, or “have/has” should be construed as designating that there are such features, integers, steps, operations, components, parts, and/or combinations thereof, not to exclude the presence or possibility of adding of one or more of other features, integers, steps, operations, components, parts, and/or combinations thereof. This detailed description is provided in order to meet statutory requirements. However, this description is not intended to limit the scope of the invention described herein. Rather, the claimed subject matter may be embodied in different ways, to include different steps, different combinations of steps, different elements, and/or different combinations of elements, similar or equivalent to those described in this disclosure, and in conjunction with other present or future technologies. The examples herein are intended in all respects to be illustrative rather than restrictive. In this sense, alternative examples or implementations can become apparent to those of ordinary skill in the art to which the present subject matter pertains without departing from the scope hereof.

Claims

1. A vertical-feed hydration system, the system comprising: a venting cap assembly configured to couple to a hydration vessel comprising an interior volume that defines a reservoir; a removable straw coupled to the venting cap assembly, wherein the removable straw is configured to extend into the reservoir; a fluid delivery tube having a first end coupled to the venting cap assembly, wherein the venting cap assembly comprises a first interior pathway defining a first flow path between the removable straw and the fluid delivery tube; and a mouthpiece comprising a flow control valve, the mouthpiece coupled to a second end of the fluid delivery tube, wherein the mouthpiece comprises a second interior pathway defining a second flow path between the fluid delivery tube and a drinking port of the mouthpiece; and wherein the venting cap assembly comprises an equalization valve configured to open in response to a suction applied to the mouthpiece, to pass an equalizing air into the reservoir from an environment external to the hydration vessel.

2. The system of claim 1, wherein the flow control valve comprises at least one of a position-

latching valve, a push-pull valve, a dial-controlled valve, a lever-controlled valve, or a button-activated valve.

3. The system of claim 1, wherein the venting cap assembly further comprises: a vessel connector configured to fasten the venting cap assembly to the hydration vessel; and a hose connector coupled to the vessel connector by a fastening mechanism, wherein the fluid delivery tube is coupled to the hose connector.

4. The system of claim 3, wherein the equalization valve is comprised within the vessel connector.

5. The system of claim 3, wherein the hose connector is coupled to the vessel connector by at least one of a quick release fastening mechanism, a threaded fastening mechanism, or a twist-to-lock fastening mechanism.

6. The system of claim 1, wherein the venting cap assembly further comprises: a vessel connector configured to fasten the venting cap assembly to the hydration vessel; and a hose connector coupled to the vessel connector by a fastening mechanism, wherein the fluid delivery tube is coupled to the hose connector; wherein the vessel connector comprises a reservoir sealing valve configured to operate from an open position to a closed position in response to a disconnection of the hose connector from the vessel connector.

7. The system of claim 6, wherein the reservoir sealing valve comprises a spring-loaded poppet valve.

8. The system of claim 1, wherein the system further comprises the hydration vessel; wherein the hydration vessel comprises an insulated hydration vessel that thermally insulates the reservoir based on a double-walled housing that includes an outer wall, an inner wall, and an insulating medium between the outer wall and the inner wall.

9. The system of claim 8, wherein the hydration vessel comprises at least one of: a stainless steel alloy material, a titanium alloy, a ceramic material, and a composite material.

10. A vertical-feed cap assembly for a hydration system, the cap assembly comprising: a vessel connector comprising: a first fastening mechanism configured to fasten to a hydration vessel, the hydration vessel comprising an interior volume that defines a reservoir; a second fastening mechanism configured to fasten a straw to the vessel connector, and support the straw in a position suspended above a bottom of the reservoir; and an equalization valve configured to open to pass an equalizing air into the reservoir from an environment external to the hydration vessel, in response to a negative pressure within the reservoir; and a removable hose connector comprising: a hose connector port configured to a fluid delivery tube; and a third fastening mechanism configured to fasten the removable hose connector to the vessel connector; wherein the vessel connector and the removable hose connector define an internal flow path for fluid flow between the second fastening mechanism and the hose connector port.

11. The cap assembly of claim 10, wherein the third fastening mechanism fastens the removable hose connector to the vessel connector based on at least one of a quick release fastening mechanism, a threaded fastening mechanism, or a twist-to-lock fastening mechanism.

12. The cap assembly of claim 10, further comprising a reservoir sealing valve configured to operate from an open position to a closed position that seals the reservoir in response to a disconnection of the removable hose connector from the vessel connector.

13. The cap assembly of claim 12, wherein the removable hose connector comprises a valve actuator that depresses a stem of the reservoir sealing valve to open the reservoir sealing valve, when the removable hose connector is fastened to the vessel connector.

14. The cap assembly of claim 10, the second fastening mechanism couples to the straw based on a keyed twist-to-lock fastening mechanism.

15. A vertical-feed hydration system, the system comprising: an insulated hydration vessel comprising an interior volume that defines a reservoir; a venting cap assembly configured to couple to the insulated hydration vessel; a straw coupled to the venting cap assembly, wherein the straw is configured to extend from the venting cap assembly into the reservoir; a fluid delivery tube having

a first end coupled to the venting cap assembly, wherein the venting cap assembly comprises a first interior pathway defining a first flow path between the straw and the fluid delivery tube; and a mouthpiece comprising a flow control valve and a drinking port, wherein the mouthpiece is coupled to a second end of the fluid delivery tube, wherein the venting cap assembly comprises a second interior pathway defining a second flow path between the fluid delivery tube and the drinking port; wherein the system is configured to draw a fluid vertically from the reservoir through the straw and the fluid delivery tube to the mouthpiece in response to a suction applied to the mouthpiece; and wherein the system is configured to drain the fluid from the fluid delivery tube in response to removal of the suction applied to the mouthpiece.

16. The system of claim 15, wherein the venting cap assembly comprises an equalization valve configured to open in response to the suction applied to the mouthpiece, to pass an equalizing air into the reservoir from an environment external to the insulated hydration vessel.

17. The system of claim 15, wherein the flow control valve comprises at least one of a position-latching valve, a push-pull valve, a dial-controlled valve, a lever-controlled valve, or a button-activated valve.

18. The system of claim 15, wherein the insulated hydration vessel is configured to thermally insulate the reservoir based on a double-walled housing that includes an outer wall, an inner wall, and an insulating medium between the outer wall and the inner wall.

19. The system of claim 15, wherein the insulated hydration vessel comprises at least one of: a stainless steel alloy material, a titanium alloy, a ceramic material, and a composite material.

20. The system of claim 15, wherein the venting cap assembly further comprises: a vessel connector configured to fasten the venting cap assembly to the insulated hydration vessel; and a removable hose connector coupled to the vessel connector by a fastening mechanism, wherein the fluid delivery tube is coupled to the removable hose connector; and wherein the vessel connector comprises a reservoir sealing valve configured to operate from an open position to a closed position in response to a disconnection of the removable hose connector from the vessel connector.
